

SAMINATHAN M

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Professional Summary

AI & Data Science undergraduate (SRMIST) with practical experience in machine learning pipelines, deep learning (NLP/CV), and full-stack development. Proficient in Python and TensorFlow, aiming to apply predictive modeling and analytical skills to real-world AI engineering challenges.

Education

B.Tech – Artificial Intelligence & Data Science(Pursuing) 2024 – 2028

SRMIST – SRM Institute of Science and Technology, Tiruchirappalli

Technical Skills

AI & Data Science: Python, TensorFlow, Scikit-Learn, Pandas, NLP, Data Visualization

Web / Full-Stack: React.js, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Firebase, HTML, CSS, JavaScript

DevOps & Tools: Git, GitHub, Docker, VS Code, SQL, Streamlit, Flask

Internship

Machine Learning Intern Jun 2026 – Jul 2026

Accent Techno Soft

- **Engineered** and optimized NLP and Computer Vision training pipelines using Python and TensorFlow, reducing model inference latency by **18%**.
- **Cleaned and structured** 15+ complex datasets exceeding **100k+ records** using Pandas and NumPy, cutting data preparation time by **25%** and ensuring zero null leakage for neural network training.
- **Implemented** neural network classification models and backpropagation routines across **10+ code reviews**, boosting baseline validation accuracy by **14%**.

Projects

NGO Data Pipeline Engine 2025 – 2026

Tech Stack: SQL · R · Python · Pandas

- Developed an end-to-end data pipeline designed to identify and track at-risk students for Non-Governmental Organizations.
- Processed comprehensive multi-variable student datasets, optimizing query runtime and delivering automated, actionable visual reports for stakeholders to improve outreach targeting.

Zero-Trust AI Security Pipeline 2025–2026

Tech Stack: Python · FastAPI · Scikit-Learn · Docker · JWT · Cryptography

- Architected a Zero-Trust machine learning pipeline implementing continuous verification, role-based access controls (RBAC), and mutual TLS (mTLS) for microservice-to-model communications.
- Implemented real-time inference guardrails and input anomaly detection algorithms to sanitize payloads, mitigating adversarial injection attacks and data poisoning with 99% validation accuracy.

Certifications

- Google Data Analytics Professional Certificate — Google / Coursera
- AWS Cloud Practitioner Essentials — Amazon Web Services (AWS)
- Introduction to SageMaker Unified Studio — Amazon Web Services (AWS)
- Building Language Models on AWS — Amazon Web Services (AWS)